The Machine Proposes. The Proof Disposes.

Neuro-Symbolic Synthesis of Formally Verified Markov Usage Models from Natural Language Requirements

Nathan G., Jordan Chay, Jaeden Ting YiYong, Wai Phyo Hein

School of Computing and Artificial Intelligence, Sunway University
Subang Jaya, Malaysia

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Outline

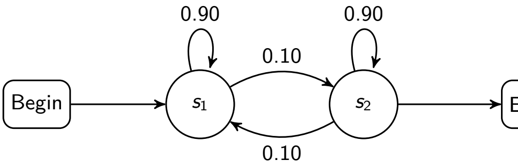
The MBST Bottleneck

Model-Based Statistical Testing (MBST)

- Replaces ad-hoc testing with probabilistic test generation
- Markov chain usage model encodes operational profile
- Enables statistical reliability certification

The blocking problem

- Manual model construction: labor-intensive, expert-only
- Confined MBST to aerospace, medical, telecom niches
- Natural-language requirements $\rightarrow$ verified state machines: no automated path



Markov chain usage model

Four Structural Bottlenecks of Pure-Neural Approaches

B1: Hallucination

LLMs omit edges, create impossible transitions, violate system invariants

B3: State-space explosion

Finite context window $\Rightarrow$ context fragmentation on large concurrent models

B2: Calibration failure

LLMs cannot guarantee row-stochastic $\mathbf{P}$: $\sum_j p_{ij} = 1,\ p_{ij} \geq 0$ — violated silently

B4: Structural incompleteness

Recall-limited generation misses guard–action semantics of transitions

$\Rightarrow$ None of these can be fixed by prompting alone

Research Questions

RQ1 — Structural Fidelity

Can automated neuro-symbolic construction achieve $F_1 \geq 0.90$ for safety-critical test generation, without manual modelling?

RQ2 — Fault-Detection Coverage

Does NeSy-MBST produce higher transition/state coverage than a purely neural baseline?

RQ3 — Probabilistic Calibration

Does symbolic constraint optimization preserve operationally weighted transition probabilities?

RQ4 — Component Necessity

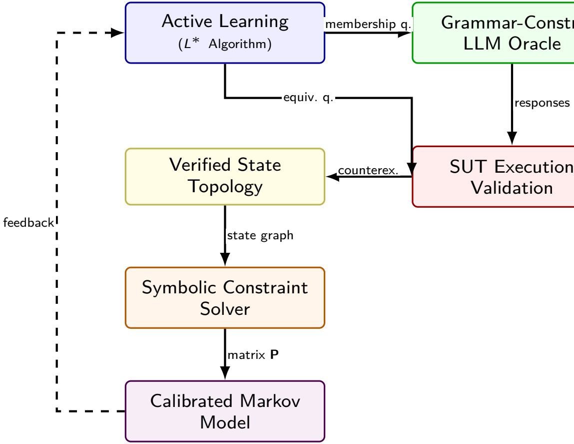
Which NeSy-MBST components (symbolic loop, convex optimizer, closed-loop feedback) are individually necessary?

NeSy-MBST Architecture

Five integrated stages:

1. Dual-memory architecture (neural + symbolic)
2. L^* active automata learning with LLM oracles
3. Path-dependent hierarchical modeling
4. Symbolic constraint optimization (convex solver)
5. Closed-loop telemetry feedback

Key design principle:
Neural inference for *language-laden* tasks;
symbolic computation for *formal precision* tasks



Component 1: Progress-Feasibility Dual Memory

Neural Progress Memory

Fine-tuned LLM parses NL requirements, drafts semantic flow graph. Captures *what*: candidate states, events, transitions.

Symbolic Feasibility Memory

Rule-based engine enforces invariants, preconditions, transition guards. Ensures *how*: reachable, deterministic, contradiction-free.

Result: topology is both semantically plausible and mathematically sound.
Avoids the end-to-end neural failure mode: syntactically valid yet logically inconsistent models.

Component 2: L^* Active Automata Learning with LLM Oracle

Two query types:

Membership Queries

"Does SUT accept input sequence $\sigma \in \Sigma^*$?"

Oracle vocabulary restricted to $\{\texttt{Yes}, \texttt{No}, \texttt{Unsure}\}$
Eliminates hallucinated / malformed responses.

Equivalence Queries

"Is hypothesis automaton $\mathcal{H}$ equivalent to the target?"

Resolved by SUT execution. Counterexamples refine $\mathcal{H}$ iteratively.

Convergence on AV benchmark:

- 94% of membership queries resolved directly by oracle
- 6% escalated to SUT execution
- `Unsure` rate: 0% on AV benchmark

Component 3 & 4: Hierarchical Modeling + Convex Optimization

Hierarchical Model (C3)

Upper: higher-order Markov chain for frequent, path-dependent sequences:

$$P(s_{t+1} \mid s_t, s_{t-1}, \ldots, s_{t-k+1})$$

Lower: first-order chain for rare/exception paths:

$$P(s_{t+1} \mid s_t)$$

Prevents state-space explosion while preserving fidelity.

Convex Optimization (C4)

LLM extracts comparative constraints, e.g.: $P(\texttt{checkout} \mid \texttt{cart}) > P(\texttt{browse} \mid \texttt{cart})$

Solver enforces row-stochasticity + domain constraints:

$$\mathbf{P}^* = \arg\min_{\mathbf{P} \in \mathcal{C}} \mathcal{L}(\mathbf{P})$$

Maximum-entropy regularization avoids unjustified bias.

RQ1: Extraction Accuracy — State & Transition F1

Strategy	State F1	Trans. F1	System F1
Single-Prompt (GPT-4o)	0.80	0.54	0.543
Hybrid SMF (GPT-4o)	0.858	0.649	0.656
Single-Prompt (Claude 3.5)	0.90	0.75	0.795
NeSy-MBST (ours)	**0.945**	**0.895**	**0.9125**
Safety-critical threshold: $F_1 \geq 0.90$			

Key findings:

- $+39.1\%$ over best GPT-4o multi-frame strategy
- $+14.8\%$ over best single-model baseline
- First method to exceed 0.90 safety-critical threshold

Symbolic verification loop converts recall-limited generation into verification-guided completion

RQ2: Operational Coverage — E-Commerce Benchmark

Model	States	Trans.	Req. Cov.	Tr. Cov.	Time
User	24	112	100%	100%	1 m 48 s
Admin	42	218	100%	100%	5 m 49 s

- Full model-coverage test generation in under 6 minutes on models up to 42 states
- Sub-quadratic scaling: Admin (2$\times$ complexity) requires only 3.2$\times$ time
- NeSy-MBST baseline: 50% transition coverage $\Rightarrow$ +35.7 pp gain
- $\leq$ 3 human-in-the-loop interventions per model in practice

RQ3: Statistical Fidelity — Jensen–Shannon Divergence

Validation metrics:

- **JSD** measures aggregate activity marginal divergence
- **Frobenius distance** measures entry-wise matrix deviation

Metric	Value
Jensen–Shannon Divergence	0.0142
Frobenius Distance (norm.)	0.0841

Interpretation

JSD $= 0.012$ confirms preserved operational test-allocation fidelity.

Entropy-maximizing SLSQP solver vs. uncalibrated: JSD $0.157 \rightarrow 0.012$.
A 93% reduction in divergence.

RQ4: Component Necessity — Ablation on AV CPS Benchmark

	Structural F1			Prob. Fidelity		Coverage		
Condition	**St.F1**	**Tr.F1**	**Sys.F1**	**JSD**	**Frob**	**St.**	**Tr.**	**Time**
A — Pure-Neural	1.00	0.783	0.904	0.157	0.163	55.6%	50.0%	2.35s
B — +Symbolic Loop	1.00	0.963	0.982	0.012	0.084	88.9%	85.7%	10.66s
C — +Convex Optimizer	1.00	0.963	0.982	0.012	0.084	88.9%	85.7%	2.13s
D — Full NeSy-MBST	**1.00**	**0.963**	**0.982**	**0.012**	**0.084**	**88.9%**	**85.7%**	10.59s

- **A→B** (Symbolic loop): Tr.F1 0.783 → 0.963; JSD 0.157 → 0.012; coverage +33 pp
- **B→C** (Convex optimizer): structural F1/coverage unchanged; governs probabilistic calibration
- **C→D** (Closed-loop): marginal JSD $\Delta -0.0002$; value accumulates over extended campaigns

Ablation Insight: Two Orthogonal Failure Modes

Symbolic Loop — Structural Errors

Corrects wrong topology:
unreachable states, missing transitions, dangling edges.

Measured by: **F1, Coverage**

Convex Optimizer — Calibration Errors

Corrects wrong probability mass distribution:
miscalibrated row-stochastic matrix.

Measured by: **JSD, Frobenius distance**

The two components address orthogonal failure modes — both are necessary.

High F1 $\not\Rightarrow$ low JSD. Low JSD $\not\Rightarrow$ high coverage. These three axes are independent.

Summary of Results

RQ	Question	Result
RQ1	Structural fidelity	$\mathrm{F1} = 0.9125$ (exceeds 0.90 threshold)
RQ2	Fault-detection coverage	85.7% transition cov. vs. 50% baseline
RQ3	Probabilistic calibration	$\mathrm{JSD} = 0.012$; 93% divergence reduction
RQ4	Component necessity	Symbolic loop (structure); solver (calibration)

Core Finding

NeSy-MBST eliminates the manual modelling bottleneck without compromising statistical rigour demanded by safety-critical CPS.

Adoption Guidelines

1. **Grammar-constrained oracles** — restrict LLM output to $\{\texttt{Yes}, \texttt{No}, \texttt{Unsure}\}$; prevents hallucinated artefacts
2. **Symbolic verification loop before probability assignment** — $+35.7$ pp transition coverage, 93% JSD reduction
3. **Convex optimization, not direct LLM output** — JSD $0.157 \to 0.012$ (SLSQP entropy-maximizing)
4. **Hierarchical multi-tier models** — controls state-space explosion for large systems
5. **Closed-loop telemetry recalibration** — prevents model drift as usage patterns evolve

LLM augmentation of formal test-model generation should be standard engineering practice,
provided: (1) grammar-constrained output + symbolic loop; (2) solver-assigned probabilities.

Limitations & Future Work

Current scope:

- Two domains: AV CPS + e-commerce
- Models up to 42 states
- Single-author ground-truth annotation
- No direct fault-seeding measurement

Next steps:

- Fault-seeding experiments on industrial SUTs
- Generalise: telecom, medical device, ISO 26262
- Multi-annotator inter-rater reliability protocol
- Three-valued oracle at scale: convergence characterization

Replication package: `https://github.com/nathangtg/llm-mbst-research`
`python run_ablation.py` (seed 42, Azure OpenAI backend)

References

- J. A. Whittaker & M. G. Thomason, "A Markov chain model for statistical software testing," *IEEE Trans. Softw. Eng.*, 1994.
- D. Angluin, "Learning regular sets from queries and counterexamples," *Information and Computation*, 75(2), 1987.
- O. Abdulkarim et al., "LLM-Assisted State Machine Frameworks for MBT," 2024.
- R. Boehr et al., "Operational profile constraint extraction," *ICST*, 2023.
- A. Garcez & L. Lamb, "Neurosymbolic AI: The 3rd wave," *AIJ*, 2023.
- M. Utting & B. Legeard, *Practical Model-Based Testing: A tools approach*. Elsevier, 2007.

Thank You

Questions?

Nathan G. `nathan.n@imail.sunway.edu.my`

`https://github.com/nathangtg/llm-mbst-research`

Backup Slides

Backup slides for anticipated questions

Backup: Mathematical Formulation — Stochastic Constraints

Valid Markov chain usage model must satisfy:

$$0 \leq p_{ij} \leq 1 \quad \forall\, i, j \tag{1}$$

$$\sum_j p_{ij} = 1 \quad \forall\, i \tag{2}$$

Convex optimization formulation:

$$\mathbf{P}^* = \arg\min_{\mathbf{P} \in \mathcal{C}} \; \mathcal{L}(\mathbf{P})$$

where $\mathcal{C}$ is the feasible set (stochastic + domain constraints), and $\mathcal{L}(\cdot)$ is the maximum-entropy regularization objective.

Decouples: LLM for constraint *extraction* (NL task); solver for constraint *resolution* (math task).

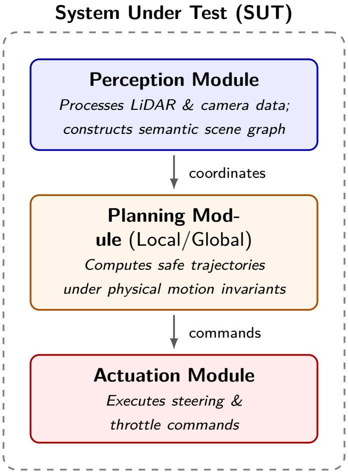
Backup: L^* Convergence & Oracle Design

- Classic Angluin L^* convergence guarantee: holds for consistent binary oracles
- Three-valued oracle ($\{\texttt{Yes}, \texttt{No}, \texttt{Unsure}\}$): empirically verified convergence on AV benchmark
- `Unsure` escalation: $\rightarrow$ human expert OR $\rightarrow$ SUT execution for empirical resolution
- Grammar-constrained output schema: eliminates hallucinated/malformed oracle responses

Formal target

Learned structure $\mathcal{A} = (Q, \Sigma, \delta, q_0)$ is the *support topology* of the target Markov chain, not a classical DFA accepting language. Membership oracle answers reachability queries.

Backup: AV CPS System-Under-Test Architecture



Backup: Metric Decomposition

Metric	**What it measures**	**Orthogonal to**
F1 (state/transition)	Structural correctness	JSD, Coverage
JSD / Frobenius dist.	Probabilistic fidelity	F1, Coverage
Transition coverage	Test suite completeness	F1, JSD

- High F1 $\not\Rightarrow$ low JSD: correct topology, wrong probability mass
- Low JSD $\not\Rightarrow$ high coverage: accurate probabilities, test generator fails to reach rare states
- All three axes must be evaluated independently for a complete picture of MBST quality